

27 2026-04-13 | Week 13 | Lecture 27

27.1 Confidence intervals

Section 7.1

Definition 126 (Confidence Interval). An interval $(\hat{p} - \epsilon, \hat{p} + \epsilon)$ is called **95% confidence interval** for the unknown parameter p if $\epsilon > 0$ is chosen large enough that

$$\mathbb{P}[|\hat{p} - p| < \epsilon] \geq .95 \quad (18)$$

In words, the interval contains the true parameter with probability at least .95.

Example 127. Can we come up with a 95% confidence interval for our broccoli poll? We already have $\hat{p} = .2$. But we don't yet know how big ϵ needs to be. (Of course we want to pick the smallest ϵ possible, as that would give us a more informative confidence interval). To figure out how big ϵ must be, we'll have to use the central limit theorem again.

Formally, let

$$\hat{p} = \frac{X_1 + \dots + X_n}{n}$$

where

$$X_i = \begin{cases} 1 & : \text{i-th sampled person likes broccoli} \\ 0 & : \text{they don't} \end{cases}$$

By the central limit theorem,

$$\hat{p} \approx p + \frac{\sigma}{\sqrt{n}}Z \quad (19)$$

where $Z \sim \mathcal{N}(0, 1)$ and $\sigma = \sqrt{\text{Var}(X_i)} = \sqrt{p(1-p)}$. We also don't know what σ is, since it depends on the unknown parameter p , but we can approximate it:

$$\sigma = \sqrt{\text{Var}(X_i)} = \sqrt{p(1-p)} \approx \sqrt{\hat{p}(1-\hat{p})} = \sqrt{(.2)(.8)} = \sqrt{.16} = .4 \quad (20)$$

By Eq. (19),

$$\begin{aligned} |\hat{p} - p| &= \frac{\sigma}{\sqrt{n}}|Z| \\ &= \frac{.4}{10}|Z| \\ &= .04|Z| \end{aligned}$$

Therefore

$$\begin{aligned} \mathbb{P}[|\hat{p} - p| \leq \epsilon] &= \mathbb{P} [.04|Z| \leq \epsilon] \\ &= \mathbb{P}[|Z| \leq 25\epsilon] \\ &= \mathbb{P}[-25\epsilon \leq Z \leq 25\epsilon] \end{aligned}$$

By the empirical rule, $\mathbb{P}[-2 \leq Z \leq 2] = .95$, so we need $25\epsilon = 2$, or $\epsilon = .08$.

Therefore the 95% confidence interval is the interval

$$(\hat{p} - \epsilon, \hat{p} + \epsilon) = (.12, .28).$$

We conclude that with 95% confidence, the true proportion of people in Hawaii who like broccoli is between 12% and 28%. Our point estimate is 0.2, and our **margin of error** is 0.08.

If we had sampled more people, our margin of error would be smaller. How many people would we have to sample to reduce the margin of error to 0.05? What about to 0.01?

End of Example 127. $\square$